Genomic Selection Decouples Unfavorable Linkage Between Kernel Discoloration Resistance and Low Grain Protein Content in Malting Barley

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The NDSU Barley Breeding Program successfully integrated a desirable low-protein QTL on chromosome 6H; however, this resulted in co-inheritance of an unfavorable QTL for poor resistance to kernel discoloration (KD), creating a negative linkage that complicates traditional selection. Kernel discoloration is a critical quality defect in malting barley (*Hordeum vulgare* L.), primarily influenced by genotype, environment, and fungal infection. Meeting malting specifications requires maintaining both KD resistance and low grain protein content (GPC), creating a dual-trait challenge for effective selection. To enable simultaneous, efficient improvement of these polygenic, negatively correlated traits, we developed and evaluated genomic selection (GS) models. This study used 205 cultivars and advanced breeding lines grown across three North Dakota environments over two years as a training population. An independent testing population of 22 lines was sourced from late-stage yield trials. All lines were phenotyped for KD using a color spectrophotometer and for GPC using near-infrared reflectance spectroscopy and were genotyped with the 50K Illumina iSelect array. Sixteen GS models were evaluated to determine the optimal model for each trait. The optimal model for KD resistance was the Reproducing Kernel Hilbert Space (RKHS) model, achieving a predictive ability of 0.456. For GPC, the optimal model was Random Forest reaching a predictive accuracy of 0.570. While the GPC model performed robustly across low-to-moderate trait values, both models showed reduced predictive power at the extreme ends of the phenotypic distribution. These results demonstrate that GS is a promising framework to accelerate malting barley improvement, though further optimization is needed to fully resolve the complex genetic trade-offs and maximize genetic gain.

Introduction

- ✓ Kernel discoloration is associated with poor quality grain, affecting farmers' gain, up to $1.00 per bushel in 2022.
- ✓ QTLs for discoloration have been identified to be co-located with the low-protein QTL on chromosome 6H inherited from the cultivar 'Karl'.
- ✓ 'Karl' has been used by many breeding programs to reduce the grain protein content to acceptable levels (i.e., <13.0%).

Figure 1: a) brightly colored barley kernels; b) differences in kernel discoloration across different barley varieties.

Materials and Methods

- ✓ A training population of 205 lines from the NDSU barley breeding program was evaluated in three locations and two years (2023-2024) in ND.
- ✓ An independent testing population of 22 lines from late-stage yield trials was also grown and evaluated over two years and multiple locations.
- ✓ Kernel color measured via spectrophotometer (L-value) and grain protein assessed via NIR.

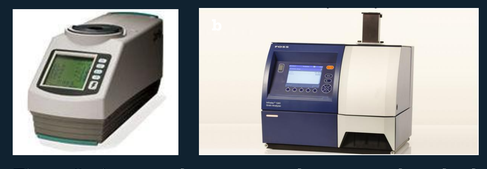

Figure 2. a) spectrophotometer used to measure kernel color; b) Near Infrared –NIR use to quantify grain protein.

- ✓ Best linear unbiased estimators (BLUEs) were calculated for individual locations.
- ✓ BLUEs were then used to calculate Best Linear Unbiased Predictors –BLUPs across locations analyses, respectively.
- ✓ Genotyping: 50K Illumina Infinium iSelect chip array.
- ✓ SNP data QAQC (MAF ≤ 0.05, missing data ≥ 0.10).
- ✓ Genomic selection model development and evaluation was done using PredictPro which tests 16 models

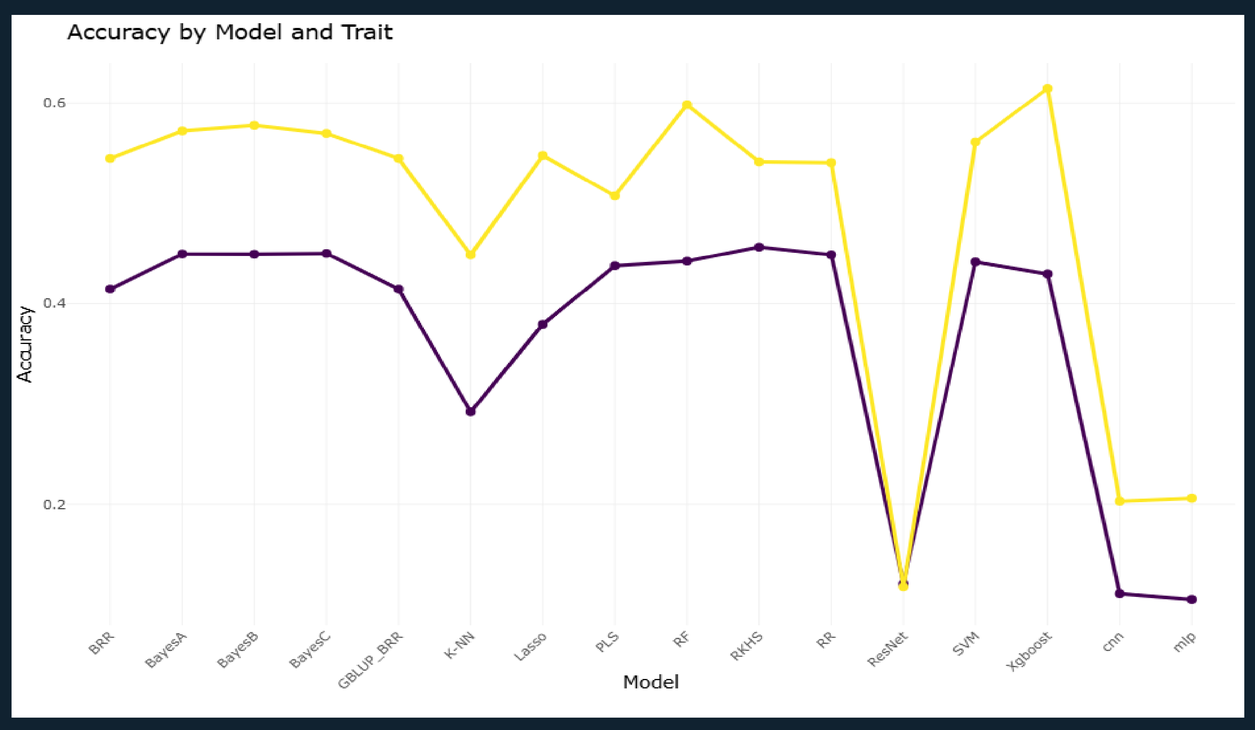

Figure 3: Performance evaluation for all 16 models tested using PredictPro for each trait.

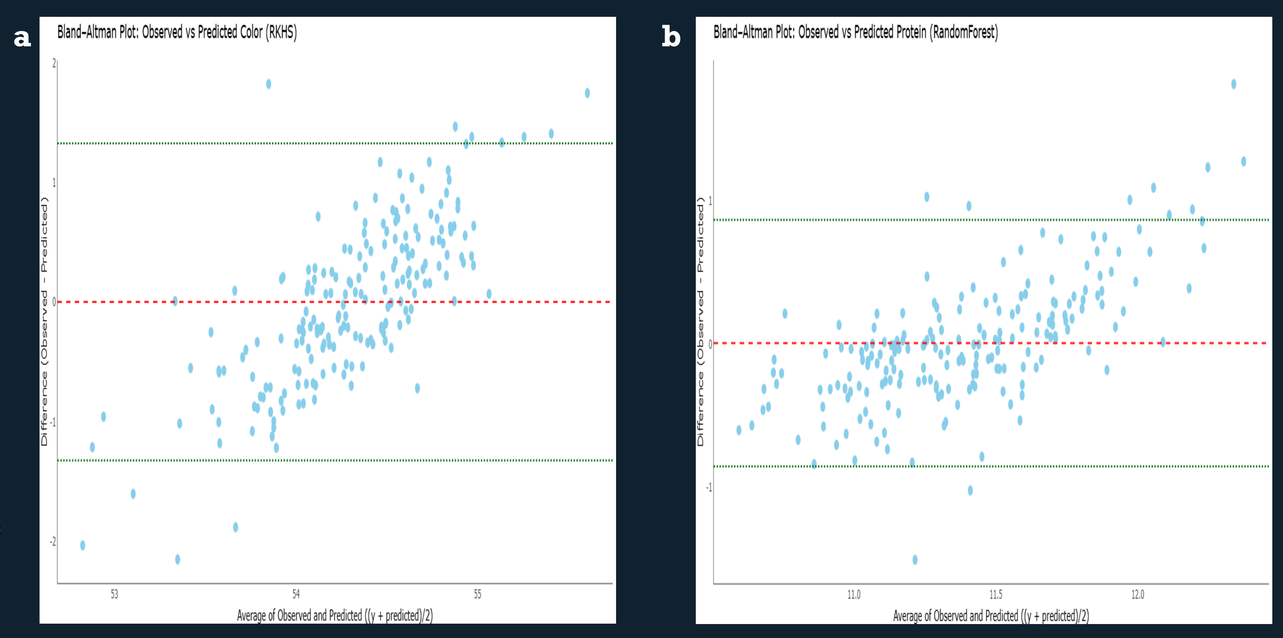

Figure 4: a) Observed versus predicted values for the Reproducing Kernel Hilbert Space Model for kernel color. b) Observed versus predicted values for the Random Forest Model for grain protein content.

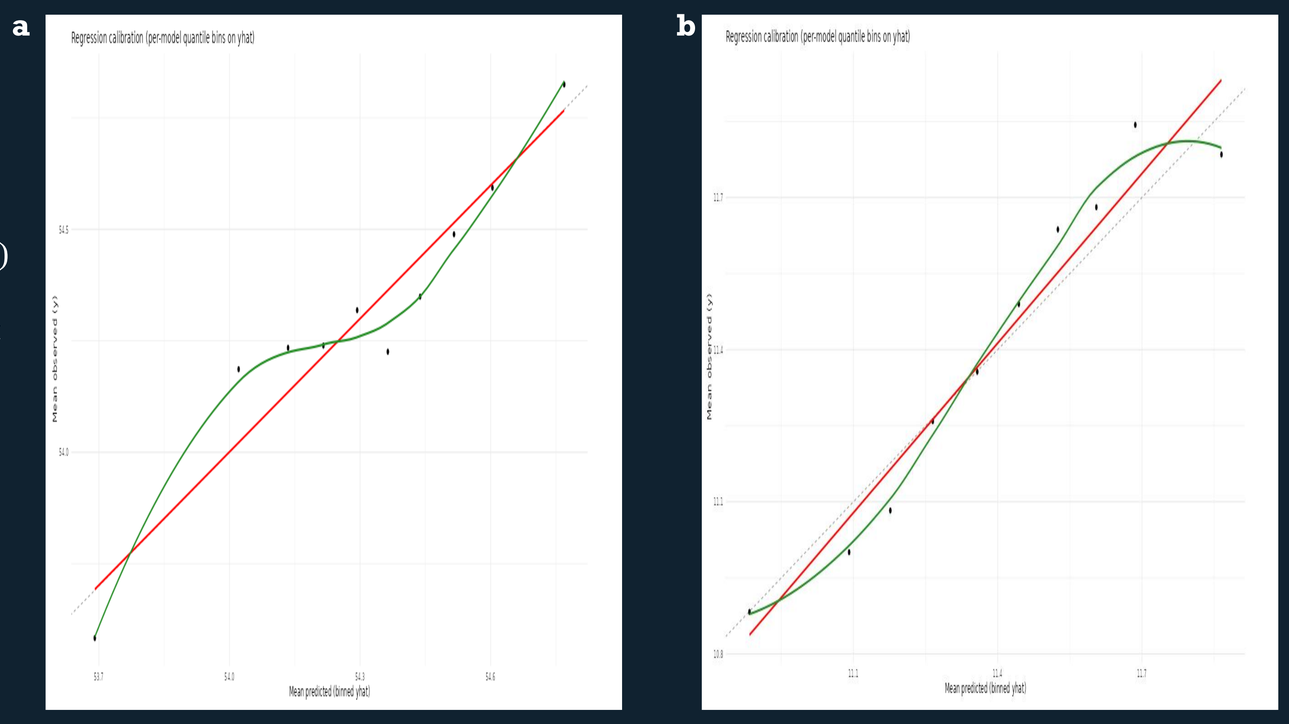

Figure 5: a) Regression calibration plot of the Reproducing Kernel Hilbert Space Model for kernel color. b) Regression calibration plot of the Random Forest Model for grain protein content.

Results and Discussion

- ✓ The Reproducing Kernel Hilbert Space Model had the best predictive ability for kernel discoloration at 0.456
- ✓ The Random Forest Model preformed best for grain protein content with a predictive ability of 0.570
- ✓ The XGBoost model appears to struggle at predicting lines that will have higher than moderate grain protein content
- ✓ Both models struggle at predicting lines with extreme trait values indicating more lines in these regions need to be included in the training model
- ✓ Both models demonstrate promise to minimize the genetic trade-offs between the two traits and speed up the process of developing cultivars with acceptable performance for both traits

Future Directions

- ✓ Collect more grain protein and kernel color data (2025) to determine if genetic haplotypes for each trait can be identified.
- ✓ Improve genomic selection models by adding in more phenotype data
- ✓ Develop and evaluate a larger set of lines for the testing population
- ✓ Additionally, we will develop a high-throughput model using an RGB camera that can predict L-values

Acknowledgements

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